# Bartlomiej Kus

Solution Architect · Principal-Level Platform & Full-Stack Engineer

Edmonton, Alberta, Canada · bartekus@gmail.com · 778-232-6992
bartekus.com · github.com/bartekus · linkedin.com/in/bartekus
statecraft.ing · github.com/statecrafting

## PROFILE

Full-stack and platform engineer with 10+ years across startups, scaleups, and enterprise and government environments. Works as a collaborative member of cross-functional, multidisciplinary teams while operating at platform/principal scope; modernizing legacy systems, designing scalable backend services and distributed-system patterns in Rust, Go, Node.js, and TypeScript, and building React/TypeScript and Vue/TypeScript front-ends. Deep experience in digital identity (OIDC, OAuth2, JWT, SSI/DID), Git-based CI/CD and infrastructure-as-code, and AI-accelerated engineering; including authoring deterministic, hash-verifiable governance tooling in Rust and building governed agentic workflows with an agentic harness and MCP, not only consuming AI tools. Pairs hands-on delivery with the mentorship, design records, and stakeholder communication that keep teams aligned.

## PROFESSIONAL EXPERIENCE

**Solution Architect · Government of Alberta, Technology & Innovation** Jan 2026 – Jun 2026
*Edmonton, Alberta · Full-time (term)*

- Embedded in an AI Delivery & Enablement product team delivering agentic-AI pipeline architecture and governed platform delivery, collaborating across platform, product, and ministry stakeholders.
- Designed and operated systems that connected enterprise source systems with consuming applications, enforcing integration contracts, architectural standards, and production quality across the delivery lifecycle.
- Applied contract-anchored governance and AI-assisted engineering practices to improve delivery consistency, traceability, and auditability.

**Principal Platform Engineer · ATB Financial / ATB Ventures** Apr 2021 – Feb 2025
*Edmonton, Alberta · Full-time*

- Mentored and coached engineers across backend, frontend, and platform domains (within my own team and adjacent teams), leading architectural discussions, design reviews, and code reviews to build a collaborative, high-quality engineering culture.
- Authored RFCs and architecture decision records (ADRs) and fronted them with presentations and technical showcases that aligned technical and non-technical stakeholders around platform direction.
- Led engineering for the Oliu.id digital-identity platform, supporting a multi-service architecture and improving system reliability, developer workflows, and user experience.
- Led a major refactor of the Oliu identity backend, improving modularity, maintainability, and integration consistency; implemented platform-level API patterns to standardize identity-verification (OIDC) integrations and reduce partner onboarding friction.
- Modernized legacy UIs into React/TypeScript progressive web applications integrated with backend services, and contributed to the Insight Engine redesign, improving processing workflows and performance.
- Supported Kubernetes deployments and multi-environment configuration; delivered containerized services and AWS Lambda functions on partner and prospective-partner engagements; automated local development, environment provisioning, and system initialization to reduce onboarding time and errors.
- Leveraged AI-assisted development tools (Claude, Cursor, Copilot) to accelerate implementation, testing, and documentation.

**Senior Software Engineer · Magnify Digital** Apr 2020 – Apr 2021
*Edmonton, Alberta · Full-time*

- Led full-stack development for the platform that evolved into ScreenMiner, stabilizing core workflows and reshaping its technical foundation.
- Owned end-to-end feature delivery across frontend, backend, and DevOps using React, Node.js, and MariaDB on AWS; rebuilt legacy frontend components into a modern React architecture.
- Optimized backend and database performance, improving stability during high-traffic operations.

**Front-End Developer · Redman Technologies Inc** Oct 2019 – Mar 2020
*Edmonton, Alberta · Full-time*

- Built a multi-tenant React front-end for real-estate marketing automation with deep integration into Couchbase (NoSQL) workflows.
- Implemented React + Redux-Saga workflows for postcard automation and image-processing pipelines, including a polling and synchronization layer for high-volume image processing.

**Senior Software Engineer · Blockchain Foundry Inc** May 2018 – May 2019
*Burnaby, British Columbia · Full-time*

- Co-developed a React/Redux browser extension enabling document signing and blockchain workflows on Syscoin, with Bitcoin and Ethereum support; integrated blockchain APIs across multi-chain interactions.
- Implemented CI/CD pipelines on AWS using Jenkins, with Terraform provisioning the AWS resources, and automated local blockchain environments to streamline deployments and developer onboarding.

**Senior Software Engineer · Beanworks Solutions Inc** May 2017 – May 2018
*Vancouver, British Columbia · Full-time*

- Shipped features across a React–Backbone.js platform and built a React Native expense-capture app; automated PostgreSQL ingestion pipelines with cron, Bash, and PSQL.

**Lead Developer · Support Services Unlimited** Feb 2017 – May 2017
*Vancouver, British Columbia · Full-time*

- Modernized a legacy WebForms application and improved developer practices through architectural guidance and code reviews.

**JavaScript Developer · Semios Technologies Inc** Nov 2016 – Feb 2017
*Vancouver, British Columbia · Full-time*

- Built a redesigned Ember.js dashboard and expanded automated test coverage across unit, integration, and acceptance tests.

**JavaScript Developer · Strutta Media Inc** Aug 2016 – Nov 2016
*Vancouver, British Columbia · Full-time*

- Built end-to-end test suites for an Ember + Rails platform, improving regression detection and release confidence.

**JavaScript Developer · Assembly Co** Feb 2016 – Aug 2016
*Vancouver, British Columbia · Full-time*

- Implemented customer-rewards features in a Node.js/Angular/Ionic application and automated Gulp build workflows.

**Control Systems Technician · Modern Systems Management Ltd** Apr 2012 – May 2015
*Surrey, British Columbia · Full-time*

- Developed custom UI dashboards and Java backend extensions for HVAC and building-automation control systems (Schneider Electric EcoStruxure), integrating web technologies with Java-based backends for real-time monitoring and control.

## SELECTED PROJECTS

**spec-spine**
Open source · Apache-2.0 · statecraft.ing · github.com/statecrafting/spec-spine

A typed, hash-verifiable authority ledger over a markdown specification corpus: an installable Rust library and CLI that refuses code which drifts from its owning specification at PR time. The specification authority underneath every statecrafting repository, and it governs itself: its own coupling gate runs against its own spec corpus in CI.

- Built as three Rust crates (typed schema, core engine, thin CLI) behind a JSON-in / JSON-out facade; every verdict-rendering verb (check, compile, index, lint, couple, delta, verify) shares one exit-code contract (0 ok, 1 finding, 2 refused, 3 usage, 4 failed) and a canonical --json envelope that changes what is written, never what is decided.
- Deterministic by construction: every artifact-producing function is a pure function of (config, file contents), and a CI gate recompiles the corpus across four target triples, failing unless the emitted registry and index shard trees are byte-identical. Per-unit shard trees let parallel pull requests touch different specs without conflicting on a shared hash line.
- Released at v0.32.0 across three registries: library and CLI on crates.io (spec-spine-core, spec-spine-cli), plus prebuilt per-platform binaries that need no Rust toolchain on npm and PyPI/uvx (spec-spine).

**statecraft-cli**

Open source · Apache-2.0 · statecraft.ing/get-started · github.com/statecrafting/statecraft-cli

A local environment for governed agent work: register a repository, prepare an isolated workspace, supervise an agent through one unit of work, judge the result independently, and keep a reviewable account of what happened. One machine, one repository, no account, no hosted service.

- Acceptance is independent of the agent's claim: the suite runs over identified candidate bytes from instructions read at a trusted base revision; a zero exit code is not read as success where a structured report exists, and a check that did not run is unknown, never a pass.
- A candidate cannot enlarge its own authority: policy, verifier, hooks and acceptance instructions are read at the base, never from the candidate, and refusals are counted from the adapter's event stream where the supervised process cannot reach them.
- Specified (specs 000 to 006 ratified September 2026), implemented and tested across the Cargo workspace; publication is deliberately deferred, so it runs from a checkout and no release is claimed.

**rahi**

Open source · Apache-2.0 · statecraft.ing/registry/rahi · github.com/statecrafting/rahi

The Rust chassis for governed cells: identity (co-deployed rauthy, reached only through the application's own origin), replicated state (in-process hiqlite Raft), a hash-chained decision ledger, a deny-by-default capability kernel, an axum edge with probes, metrics and tracing, and single-container packaging with preflight, migrate, backup and restore. An application composes the crates and declares a manifest; it extends the chassis and never forks it.

- Nine crates published to crates.io through 0.7.1 (release candidate), with a reference application and a dev substrate that boots the real binary in tests.
- Exercised end to end against a digest-pinned rauthy: login, audience and scope enforcement, fresh backups, restart, and restore with the original user, note and ledger head.
- Supported topology is stated as N=1; a three-replica composition is built and awaiting operator qualification on a real cluster, and the README says so rather than implying it.

**rustev, wire-witness, aicortex**

Open source · Apache-2.0 · statecraft.ing/products · github.com/statecrafting

- **rustev** (Rust, v0.5.0 on crates.io): an embeddable engine for making, evaluating and recording decisions over partially trusted information; versioned decision definitions compile into bounded plans with explicit abstention and an evidence record per decision. It produces proposals, never grants.
- **wire-witness** (Rust, v0.1.0): captures provider wire exchanges through a child-scoped proxy and produces attributed observations with custody facts; its output is testimony and never grants authority to execute, admit evidence or decide acceptance.
- **aicortex** (Rust, specified and being built one spec per pull request): persistent, governed memory for AI clients over MCP, built as a rahi cell; recalled content is always data, and only a human-recorded promotion makes a memory instruction grade.

**Primitives: action-gate, attest-ledger, canonical-keysort-json**

Open source · Apache-2.0 · statecraft.ing/registry · github.com/statecrafting

- **action-gate** (v0.3.0): a pure, deterministic decision gate over an ordered, pluggable check registry, returning Allow, Deny or Degrade with a stable config hash.
- **attest-ledger** (v0.2.0): an append-only, hash-linked, Ed25519-signed record ledger with an independent verifier that does not trust the producer.
- **canonical-keysort-json** (v0.1.1): a deterministic key sort at the serialization boundary so record hashes agree across every crate in the dependency graph.

## TECHNICAL SKILLS

**Frontend & Mobile:** React, Vue, Next.js, Remix, React Native, TypeScript, JavaScript, HTML, CSS, TailwindCSS

**Backend & APIs:** Rust, Go, Node.js, TypeScript, Python, GraphQL, REST, microservices, distributed systems, API design

**Databases:** PostgreSQL, MySQL, MariaDB, SQL; Couchbase (NoSQL)

**Identity & Auth:** OAuth2, OpenID Connect (OIDC), JWT, Self-Sovereign Identity (SSI), Decentralized Identifiers (DID)

**Cloud & DevOps:** Kubernetes, Terraform, Docker, Docker Compose, CI/CD, Infrastructure-as-Code, Linux, Git, Bash, AWS, GCP, Azure (AKS), DigitalOcean, Hetzner

**AI & Agentic Engineering:** Agentic orchestration, agentic harness, MCP, AI workflows, RAG / structured retrieval, prompt engineering, AI-output evaluation, AI-assisted development (Claude, Cursor, Copilot)

**Release & Observability:** CycloneDX SBOMs, signed / verifiable installers, Prometheus, Grafana

**Blockchain (prior):** Syscoin, Bitcoin, Ethereum, Hyperledger

## EDUCATION

**Certificate in Web Development · Lighthouse Labs** Aug 2015 – Nov 2015
*Vancouver, British Columbia*

- Intensive program in modern JavaScript, full-stack fundamentals, and practical software engineering.

**Certificate, Computer Systems Service Technician · BCIT** Sep 2002 – Jun 2003
*Burnaby, British Columbia*

- Computer systems, networking, and administration with hands-on hardware and software troubleshooting.

## LANGUAGES

English (Fluent) · Polish (Fluent)

## REFERENCES

*“Bart is hands down one of the best developers I know. He's thorough, a fast implementer, a great listener, and an all-round pleasure to work with. Anyone lucky enough to have Bart on their team is a very lucky person.”*

- Sebastian Schepis, Chief Technologist, Nomyx

*Additional references available upon request.*